

Assignment 5 — Clustering and Initialization

k-means vs *k*-means++: The Advantages of Careful Seeding

Statistical Methods for Machine Learning · Università degli Studi di Milano · Instructor: Nicolò Cesa-Bianchi

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1. Introduction

Given n data points $x_1, \dots, x_n \in \mathbb{R}^d$ and a target cluster count k , *k*-means seeks centers $c_1, \dots, c_k$ minimizing the *inertia*:

$$\phi = \sum_{i=1}^n \min_j \|x_i - c_j\|^2. \quad (1)$$

Because ϕ is non-convex, Lloyd’s algorithm [2] only converges to a *local* minimum; which minimum depends entirely on the initial centers. This project compares two initialization strategies under controlled conditions: **random (Forgy)** — k centers drawn uniformly at random — and ***k*-means++** [1] — centers seeded sequentially with probability proportional to the squared distance to the nearest existing center (D^2 weighting). The methods are identical in every other respect.

Evaluation covers four axes across three datasets: **inertia** (internal), **ARI** and **NMI** (external, against ground-truth labels), **convergence speed** (Lloyd iterations), and **run-to-run stability**. All algorithms and metrics are implemented from scratch in NumPy; **scikit-learn** is used only to generate or load raw data.

2. Datasets and Pre-processing

Table 1: Datasets used. The assignment requires at least one synthetic and one real-world dataset; overlapping clusters and the k -sweep set are extensions.

Dataset	Type	$n \times d$	k	Role
Well-separated clusters	Synthetic (std=0.6)	500×2	5	Clean regime; ideal for <i>k</i> -means++
Overlapping clusters	Synthetic (std=1.8)	500×2	5	Weak-structure extension
Digits	Real-world, 10 digit classes	1797×64	10	Required high-dimensional dataset
Varying- k clusters	Synthetic (std=0.5 , 15 true clusters)	1000×2	2–15	Theory verification

Pre-processing choices. The synthetic cluster datasets require no preprocessing: **make_blobs** generates isotropic Gaussian clusters that are already well-scaled. The Digits dataset is standardized to zero mean and unit variance per feature before clustering, because *k*-means is not scale-invariant — features with larger absolute values dominate the Euclidean distance and therefore the objective. Features with zero variance are set to unit standard deviation to avoid division by zero.

Reproducibility. All experiments use **random_state=42**, 50 independent restarts, **max_iter=300**, and **tol=10⁻⁴** (max center shift). A critical implementation detail: the empty-cluster fallback draws from the same *seeded* generator as the rest of the run — not NumPy’s global RNG — so every step of a given **random_state** is fully reproducible.

3. Algorithms and Implementation

3.1 Theoretical background

Lloyd's algorithm alternates (i) **assignment**: $\ell(x_i) = \arg \min_j \|x_i - c_j\|^2$, and (ii) **update**: $c_j \leftarrow |C_j|^{-1} \sum_{x \in C_j} x$. Each step decreases ϕ , guaranteeing convergence to a local minimum.

k-means++ seeding. Pick c_1 uniformly; then for each subsequent center sample x with $p(x) \propto D(x)^2 = \min_{j < \ell} \|x - c_j\|^2$. This D^2 weighting spreads seeds away from existing centers at cost $O(knd)$, negligible versus Lloyd's $O(I \cdot n \cdot k \cdot d)$ total cost. Its key guarantee is:

Theorem 1 (Arthur & Vassilvitskii, 2007 [1]). $\mathbb{E}[\phi_0] \leq 8(\ln k + 2) \cdot \phi^*$, where ϕ^* is the optimal inertia and ϕ_0 is the initialization cost.

This $O(\log k)$ -competitive bound predicts that the advantage of k -means++ grows with k — verified empirically in Section 5.3.

3.2 From-scratch implementation

All code is NumPy-only. The central function is:

```
def kmeans(X, k, init='random', random_state=None, max_iter=300, tol=1e-4):
    """Returns: centers (k,d), labels (n,), inertia (float), n_iter (int)."""
    rng = np.random.default_rng(random_state) # one seeded generator per
    run
    centers = INITIALIZERS[init](X, k, rng) # pluggable init strategy
    for i in range(1, max_iter + 1):
        labels = assign_labels(X, centers) # vectorised: (n,k,d) ->
        argmin
        new_centers = update_centers(X, labels, k, rng)
        if np.max(np.linalg.norm(new_centers - centers, axis=1)) < tol:
            break
        centers = new_centers
    return centers, assign_labels(X, centers), compute_inertia(X, labels,
        centers), i
```

The INITIALIZERS registry maps 'random' to Forgey sampling and 'kmeans++' to the D^2 seeding loop; adding a new strategy requires only inserting it into the registry. Assignment uses vectorized broadcasting $(n, d) \rightarrow (n, k, d) \rightarrow \argmin_{(n,)};$ the update step re-seeds empty clusters from the seeded generator rather than the global RNG.

4. Evaluation Metrics

Inertia (ϕ , Eq. 1) is the k -means objective — lower is tighter — but it is not comparable across datasets (it scales with n , d , and feature spread) and is decoupled from true cluster quality in high dimensions. Because every dataset here has ground-truth labels we add two external metrics, both implemented from scratch via a contingency table.

Adjusted Rand Index (ARI) [3] counts concordant pairs and corrects for chance; $\text{ARI} = 1$ for a perfect partition, $\text{ARI} \approx 0$ for random agreement. **Normalized Mutual Information (NMI)** measures shared information between the predicted and true labelings, normalized by their mean entropy; $\text{NMI} = 1$ is perfect.

Near-optimal fraction $f_\tau = T^{-1} \sum_t \mathbf{1}[J_t \leq J^*(1 + \tau)]$ (we use $\tau = 0.01$) directly measures how often a method finds a near-optimal solution across restarts, making it more interpretable than the coefficient of variation ($\text{CV} = \text{std}/\text{mean}$).

Paired seeds. Both strategies are evaluated on the *same* 50-seed sequence, so any difference in outcomes is attributable to the initialization, not to luck in the restart draw.

5. Experiments and Results

5.1 Experiment 1: Well-separated clusters

Table 2: Experiment 1 — well-separated clusters ($k = 5$, 50 restarts).

Method	Mean J	Std J	CV	Near-opt	ARI	Iters
Random	1109.2	889.6	80.2%	42%	0.833	5.58
k -means++	439.4	331.8	75.5%	88%	0.966	3.10

Experiment 1: Well-separated clusters ($k=5$)

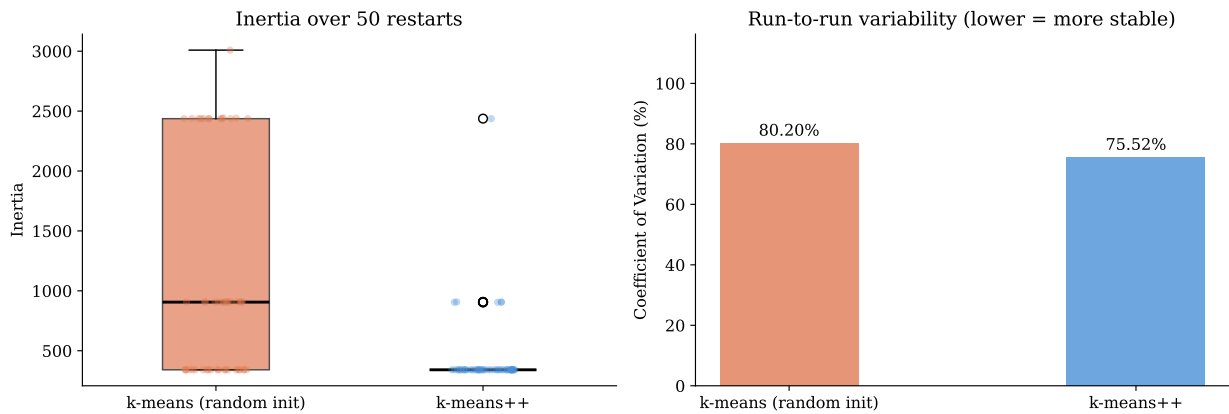


Figure 1: Inertia distribution and CV over 50 restarts — well-separated clusters. k -means++ is both lower and more consistent.

k -means++ dominates on every axis. It reduces mean inertia by **60.4%** (439 vs 1109), raises ARI from 0.833 to **0.966** and NMI from 0.919 to 0.984, and converges in **3.1 iterations** vs 5.6 — roughly half as many Lloyd steps. The near-optimal fraction rises from 42% to **88%**: random init fails to find a near-optimal solution in more than half of runs.

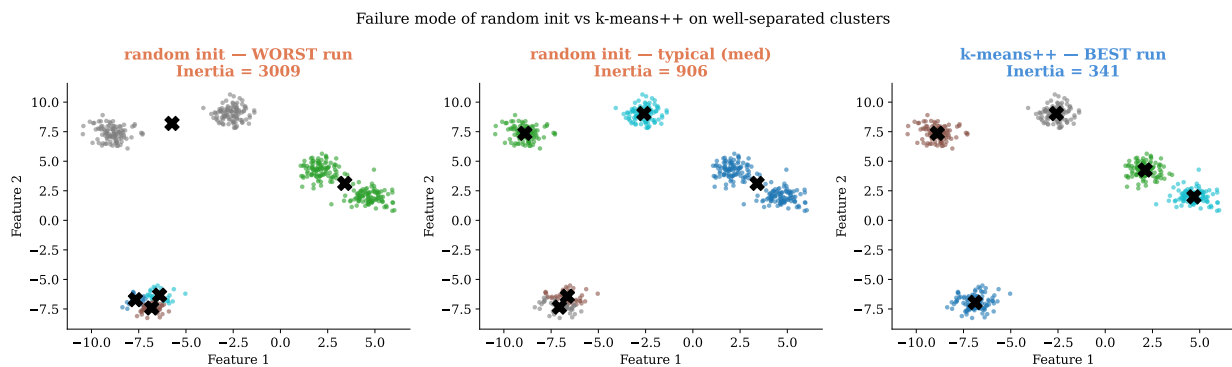


Figure 2: Failure mode of random init. Left: the worst random run — two seeds in the same true cluster, locking in a bad local minimum. Center: a typical random run. Right: the best k -means++ run. Black crosses = final centers.

Figure 2 shows the failure mechanism directly. In the worst random run two seeds landed in the same cluster, splitting it while merging two adjacent ones. Lloyd’s updates cannot undo this — once centers are placed, every assignment step is optimal *given* those centers, locking in the sub-optimal partition. k -means++ avoids this by placing one seed in each cluster via the D^2 rule.

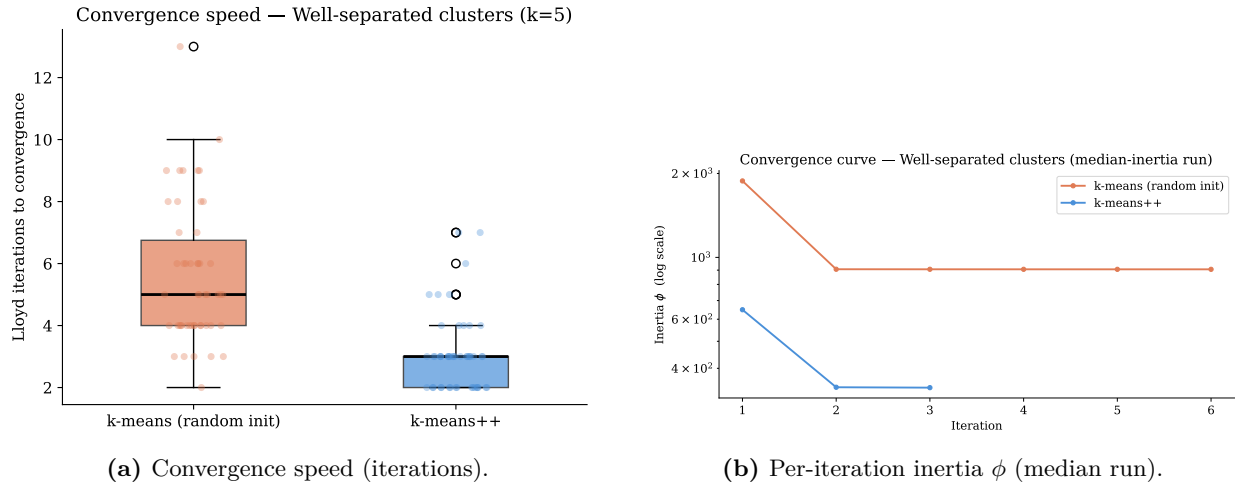


Figure 3: Left: k -means++ needs $\sim$ half the Lloyd iterations. Right: inertia ϕ decreases monotonically across Lloyd iterations; k -means++ starts from a lower ϕ at iteration 1 (seeding advantage) and converges sooner. Convergence is guaranteed because Lloyd monotonically decreases ϕ .

5.2 Experiment 2: Overlapping clusters

Table 3: Experiment 2 — overlapping clusters ($k = 5$, 50 restarts).

Method	Mean J	Std J	CV	Near-opt	ARI	Iters
Random	2915.3	423.3	14.5%	74%	0.780	11.26
k -means++	2930.8	273.6	9.3%	64%	0.768	9.42

Experiment 2: Overlapping clusters ($k=5$)

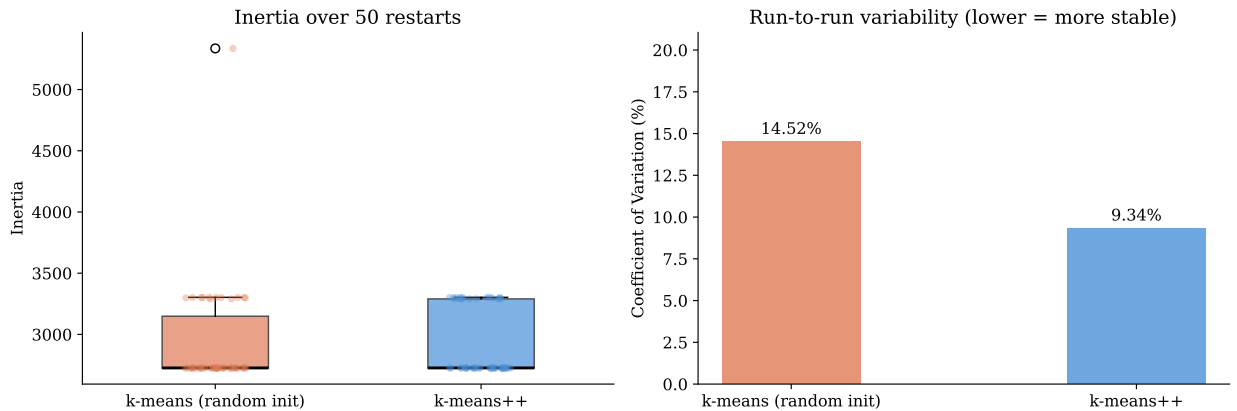


Figure 4: Overlapping clusters: similar mean inertia ($< 0.5\%$ difference), but k -means++ has noticeably lower variability (CV 9.3% vs 14.5%).

The picture changes entirely. Both methods reach nearly identical inertia ($< 0.5\%$ difference), and external quality is indistinguishable (ARI 0.780 vs 0.768, NMI 0.832 vs 0.828). k -means++ gains are now confined to **stability** (CV 9.3% vs 14.5%, a 36% reduction) and **speed** (9.4 vs 11.3 iterations). When clusters overlap substantially the objective landscape flattens — many configurations achieve similar inertia — so the starting point is less consequential for final quality. The near-optimal fraction even inverts (64% vs 74%): with a flat landscape the 1% tolerance threshold is easy to exceed, reflecting landscape ambiguity rather than inferior performance.

5.3 Experiment 3: Effect of k and link to theory

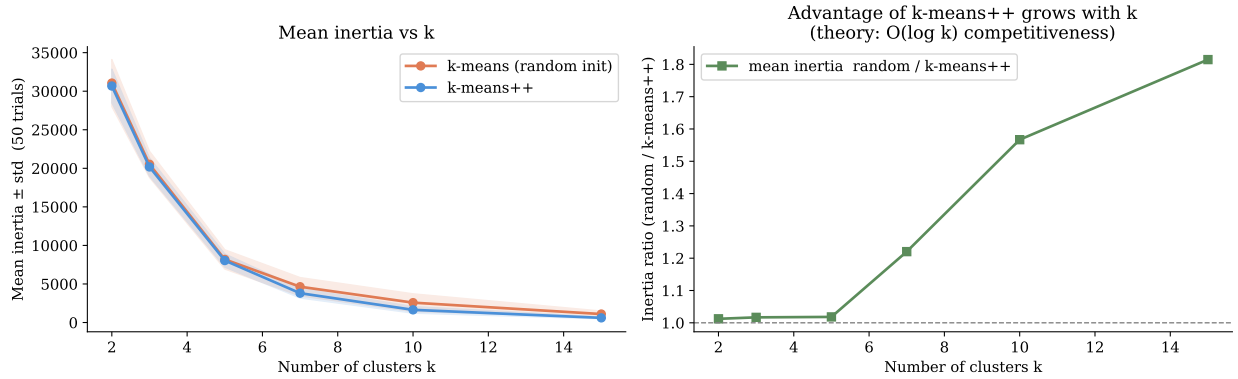
Experiment 3: Effect of k 

Figure 5: Left: mean inertia vs k . Right: random/ k -means++ inertia ratio vs k . The ratio grows monotonically from 1.01 at $k = 2$ to 1.82 at $k = 15$, tracking $\ln k$, directionally consistent with the $O(\log k)$ theory.

Table 4: Inertia ratio vs k : $\bar{J}_{\text{rand}}/\bar{J}_{++}$ grows with $\ln k$.

k	$\bar{J}_{\text{rand}}$	$\bar{J}_{++}$	Ratio	$\ln k$
2	31053	30675	1.012	0.69
3	20519	20178	1.017	1.10
5	8187	8041	1.018	1.61
7	4643	3805	1.220	1.95
10	2578	1645	1.567	2.30
15	1099	606	1.815	2.71

The advantage of k -means++ grows **monotonically** from 1.012 at $k = 2$ to **1.815** at $k = 15$, tracking $\ln k$ closely (Table 4). For small k (2–5) the advantage is negligible: with few clusters, random init rarely seeds two centers in the same cluster. At $k = 7$ the ratio jumps to 1.22 as seeding errors become frequent. This is directionally consistent with the $O(\log k)$ competitive bound: more clusters give random init more opportunities to fail, widening the gap.

5.4 Experiment 4: Real-world high-dimensional data (Digits)

Table 5: Experiment 4 — Digits ($d = 64$, $k = 10$, 50 restarts).

Method	Mean J	Std J	CV	Near-opt	ARI	Iters
Random	70,493	883	1.25%	42%	0.512	25.58
k -means++	71,381	959	1.34%	16%	0.464	25.12

Experiment 4: Digits 64-D (k=10)

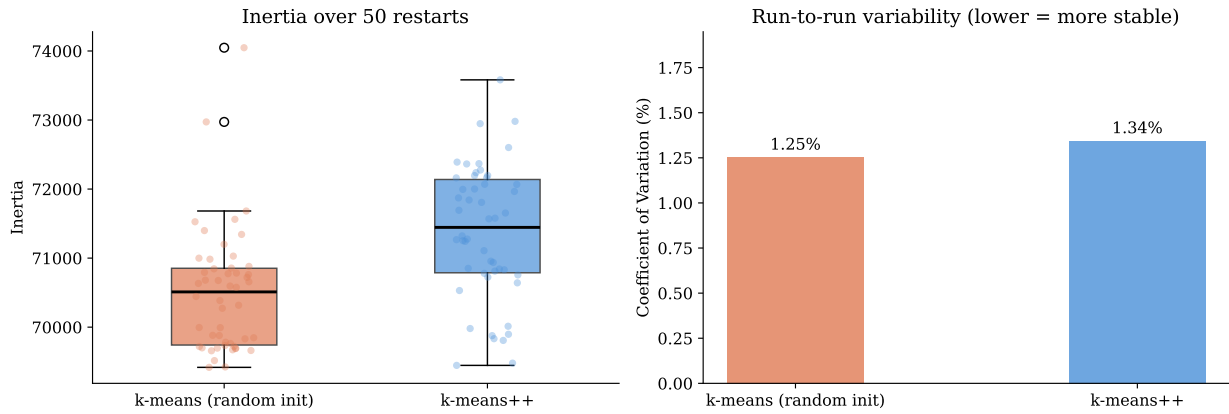


Figure 6: Digits 64-D: nearly identical, very tight inertia distributions ($CV \approx 1.3\%$ for both). The objective landscape is effectively unimodal.

The result **inverts**. k -means++ is marginally *worse*: inertia +1.3%, ARI 0.464 vs **0.512**, NMI 0.618 vs **0.656**. Three mechanisms contribute:

1. **Distance concentration.** In 64 dimensions, all pairwise distances become similar ($\max / \min \rightarrow 1$), flattening the D^2 distribution so k -means++ sampling is nearly uniform — its seeding signal vanishes.
2. **Landscape unimodality.** $CV \approx 1.3\%$ for *both* methods — $50\times$ lower than Experiment 1. The landscape has effectively one basin; initialization choice is irrelevant for final inertia.
3. **Greedy spreading on outliers.** k -means++ actively maximizes spread, which can anchor seeds on atypical digit images. Such outlier seeds attract few points, producing unbalanced clusters that score worse on ARI/NMI.

Methodological lesson. Inertia alone would suggest equivalence (both $\approx 70,000$, $CV \approx 1.3\%$). Only external metrics reveal the reversal and its direction. A study relying only on the internal objective would miss this entirely.

5.5 External validation across datasets

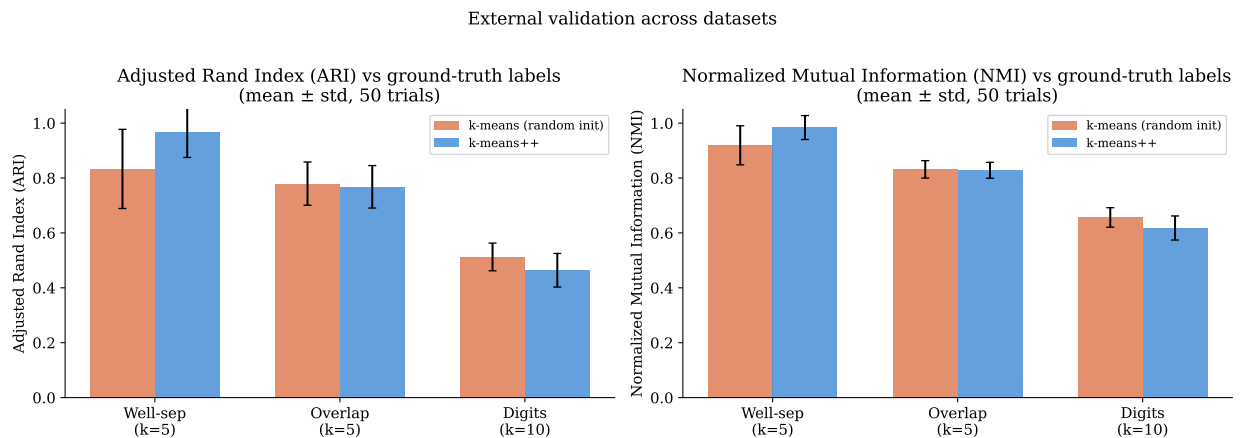


Figure 7: ARI and NMI across all three labelled datasets (mean $\pm$ 1 std, 50 restarts). k -means++ leads only on well-separated data; ties on overlapping; trails on Digits.

Figure 7 places all three regimes side by side. The pattern is unambiguous: k -means++ improves ARI by +0.13 on well-separated data, matches random on overlapping data, and trails by -0.05

on Digits. The error bars confirm that random init is most variable on well-separated data (wide) and barely variable on Digits (narrow), consistent with the landscape-structure story across all experiments.

6. Summary and Discussion

Table 6: Full cross-dataset summary (50 restarts, `random_state=42`). $\Delta J\%$: percentage reduction in mean J by k -means++; positive = better.

Dataset	$\bar{J}_r$	$\bar{J}_{++}$	$\Delta J\%$	CV_r	CV_{++}	ARI_r	ARI_{++}	NMI_r	NMI_{++}	it_r/it_{++}
Well-sep (2D, $k=5$)	1109	439	+60.4	80.2	75.5	0.833	0.966	0.919	0.984	5.6 / 3.1
Overlap (2D, $k=5$)	2915	2931	−0.5	14.5	9.3	0.780	0.768	0.832	0.828	11.3 / 9.4
Digits (64D, $k=10$)	70493	71381	−1.3	1.3	1.3	0.512	0.464	0.656	0.618	25.6 / 25.1

The experiments tell a coherent, theory-consistent story.

Finding 1 — Strong structure: k -means++ wins decisively. A 60.4% inertia reduction, ARI from 0.833 to 0.966, NMI from 0.919 to 0.984, half the iterations, and 88% near-optimal rate vs 42% for random init. The scatter plots confirm the mechanism: D^2 seeding prevents the pathological two-seeds-in-one-cluster initialization that Lloyd cannot recover from.

Finding 2 — Weak structure: stability and speed, not quality. Both methods reach the same partition quality ($ARI \approx 0.77$). k -means++ still reduces CV by 36% and cuts iterations from 11.3 to 9.4, which matters in any single-run or time-constrained setting.

Finding 3 — High dimensions: the advantage reverses. On 64-D Digits, k -means++ is marginally worse on inertia, ARI, and NMI. Distance concentration eliminates the D^2 signal; the landscape becomes near-unimodal; and greedy spreading can misplace seeds on outlier images. This negative result is only visible through external metrics.

Finding 4 — Growing advantage with k , matching theory. The inertia ratio grows from 1.01 at $k = 2$ to 1.82 at $k = 15$, tracking $\ln k$ and directionally consistent with the $O(\log k)$ competitiveness guarantee.

Visualization choices. Box-and-strip plots expose the full inertia distribution (not just mean $\pm$ std), making bimodal outcomes visible. The scatter triptych (worst/typical/best) illustrates the failure mechanism directly. The convergence curve uses the median run to avoid cherry-picking; bar charts include error bars for variability.

7. Conclusions

k -means++ is a strong default whose benefit is *conditional* on strong, low-dimensional structure and large k , matching the $O(\log k)$ theory. The key lesson is that initialization studies must pair inertia with external label-based metrics (ARI, NMI) and a stability measure; inertia alone conceals both the high-dimensional reversal and the overlapping null result.

Limitations. (i) Synthetic datasets are isotropic equal-sized Gaussians; real clusters are often non-spherical. (ii) PCA pre-processing on Digits might reduce distance concentration and restore the advantage. (iii) Convergence tolerance 10^{-4} may not fully converge on Digits; a tighter threshold would cost more iterations.

Future directions. Apply PCA before Digits clustering; compare with k -means|| [4] (parallel $O(\log n)$ -round seeding); extend to mini-batch k -means and non-Euclidean settings.

References

- [1] Arthur, D. & Vassilvitskii, S. (2007). k -means++: The advantages of careful seeding. In *Proc. ACM-SIAM SODA*, pp. 1027–1035.
- [2] Lloyd, S. P. (1982). Least squares quantization in PCM. *IEEE Trans. Inf. Theory*, 28(2):129–137.
- [3] Hubert, L. & Arabie, P. (1985). Comparing partitions. *J. Classification*, 2:193–218.
- [4] Bahmani, B. et al. (2012). Scalable k -means++. *Proc. VLDB Endowment*, 5:622–633.